JEE Mathematics: Quick Review with Grading and Practice Map - Circle (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (64 questions; a question that appears in two of these is counted once): High = 6 or more questions, Medium = 1 to 5 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Basic Geometry about Circles

Question 01

What is the relation between the length of a chord and the angle subtended by the chord at the centre?

Model Answer

Chord subtending 2θ at the centre: length $= 2r\sin\theta$; distance from centre $= r\cos\theta$. Angle at the circumference (major segment) $= \theta$.

Ex: $r = 3$, angle $\frac{2\pi}{3}$ at centre: chord $= 3\sqrt{3}$, mid-points of such chords lie on $x^2 + y^2 = \frac{9}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 20-21; J-II 22; J-M 1, 12 | DPP: 2(4); 5(7, 10); 8(11, 15) | Question Bank: 53, 56

Also used in: Sheet: J-A 18 | DPP: 7(9)

Question 02

Revise all properties mentioned in this heading.

Model Answer

Equal chords $\iff$ equal angles at centre $\iff$ equidistant from centre; nearer chord is longer. Angle at centre $= 2\times$ angle at circumference; angles in the same segment are equal; angle in a semicircle $= 90°$.

Cyclic quadrilateral: opposite angles sum to $180°$. Intersecting chords / secants: $AP{\cdot}PB = CP{\cdot}PD$. Alternate segment theorem. Tangents from a point are equal; the line to the centre bisects the angle between them and is the perpendicular bisector of the chord of contact.

Grade - Medium

Practice map, main concept: Sheet: J-II 4; J-M 34; C-I 4, 6, 12-13, 17-18, 22; C-II 18, 26 | DPP: 1(5, 12) | Question Bank: 49

Also used in: Sheet: J-I 20; C-I 15; C-II 15, 22 | DPP: 4(7)

Circle: Definition and Equation

Question 03

Define circle, its centre and radius.

Model Answer

Locus of a point in a plane whose distance from a fixed point (centre) is constant (radius).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 04
How to find the equation of a circle if the centre and radius are known?

Model Answer

$$(x-a)^2+(y-b)^2=r^2$$

Ex: centre $(1,1)$ touching $3x+4y+18=0$: $r=\frac{25}{5}=5$, circle $x^2+y^2-2x-2y-23=0$.

Grade - **Medium**
Practice map, main concept: Sheet: J-I 2; J-M 36
Also used in: Sheet: J-II 15, 51; J-M 5, 27; C-II 14 | DPP: 2(6) | Question Bank: 60, 64

Question 05
What is the general form of the equation of a circle?

Model Answer

$$x^2+y^2+2gx+2fy+c=0$$

Three independent constants g, f, c: three conditions fix a circle.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 06
How to find the centre and radius from the general equation of a circle?

Model Answer
Make the coefficients of x^2, y^2 unity. Centre $(-g,-f)$ = $(-\frac{1}{2}$coeff of x, $-\frac{1}{2}$coeff of $y)$, $r=\sqrt{g^2+f^2-c}$.
Ex: $3x^2+3y^2-5x-6y+4=0 \Rightarrow$ centre $(\frac{5}{6},1)$, $r=\frac{\sqrt{13}}{6}$.

Grade - **Medium**
Practice map, main concept: Sheet: J-M 23; C-I 20 | DPP: 1(1, 6-7, 13, 17-20) | Question Bank: 51
Also used in: Sheet: J-M 18; C-II 10 | Question Bank: 68

Question 07
What is the standard form of the equation of a circle?

Model Answer

$$x^2+y^2=r^2$$

Centre at the origin, radius r.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 08
What is the necessary and sufficient condition for the general equation of 2nd degree to represent a circle?

Model Answer
$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is a circle iff $a = b \neq 0$ and $h = 0$ (and, after dividing by a, $g^2 + f^2 - c \geq 0$ for a real circle).

Grade - Low
Practice map, main concept: DPP: 1(3)
Also used in: none

Question 09
What do you mean by point circle and imaginary circle?

Model Answer
$g^2 + f^2 - c > 0$: real circle; $= 0$: point circle (only the centre); < 0: imaginary circle (no real point).

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 24; J-A 7 | DPP: 1(3)

Question 10
How to find the equation of a circle passing through 3 given points?

Model Answer
Put the points in $x^2+y^2+2gx+2fy+c = 0$ and solve for g, f, c; or take the centre (a, b) equidistant from the points. A right angle at a point $\Rightarrow$ the opposite side is a diameter.
Ex: $(6, 0), (0, 8), (6, 8)$: centre $(3, 4)$, $r = 5$, circle $x^2 + y^2 - 6x - 8y = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-I 1; J-II 14, 25, 57; J-A 14; C-II 2 | DPP: 2(13)
Also used in: Sheet: J-II 33 | DPP: 3(6)

Question 11
How to prove that 4 given points are concyclic?

Model Answer
Find the circle through three of them and check that the fourth satisfies it (or check opposite angles supplementary / $PA \cdot PB = PC \cdot PD$).
Ex: $(3, 4), (-3, -4), (0, 5), (-5, 0)$ all lie on $x^2 + y^2 = 25$.

Grade - Medium
Practice map, main concept: Sheet: J-M 20 | DPP: 1(10) | Question Bank: 38
Also used in: DPP: 1(12)

Question 12
How to find the image (reflection) of a circle in a given line?

Model Answer
Radius is unchanged; the centre goes to its image in the line: $\dfrac{x-h}{a} = \dfrac{y-k}{b} = \dfrac{-2(ah+bk+c)}{a^2+b^2}$.
Compare with the given form of the image (make coefficients of x^2, y^2 unity first).
Ex: image of $x^2 + y^2 + 16x - 24y + 183 = 0$ (centre $(-8, 12)$, $r = 5$) in $4x + 7y + 13 = 0$: centre $(-16, -2)$, circle $x^2 + y^2 + 32x + 4y + 235 = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-M 2, 5, 18 | DPP: 1(2) | Question Bank: 40, 57, 60, 68
Also used in: none

Diametrical Form of a Circle

Question 13

Write the expression for the diametrical form of a circle.

Model Answer

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Ends of a diameter $A(x_1, y_1), B(x_2, y_2)$ ($\angle APB = 90°$). If x_1, x_2 are roots of $x^2 + px + q = 0$ and y_1, y_2 of $y^2 + ry + s = 0$, the circle is the sum of the two quadratics.

Ex: abscissae roots of $x^2+2x-a^2 = 0$, ordinates of $y^2+4y-b^2 = 0$: circle $x^2+y^2+2x+4y-a^2-b^2 = 0$.

Grade - Medium

Practice map, main concept: Sheet: C-II 22 | DPP: 1(4); 2(1-3, 5); 5(6, 13)

Also used in: Sheet: J-I 5; J-II 4, 47; J-M 2, 7; J-A 8, 17; C-I 8, 22 | DPP: 3(12); 5(19); 8(13) | Question Bank: 57

Question 14

What is the smallest circle passing through 2 given points A and B?

Model Answer

The circle on AB as diameter (centre = mid-point of AB, $r = \frac{AB}{2}$).

Ex: through $(1, 0), (0, 1)$: $x^2 + y^2 - x - y = 0$.

Grade - Low

Practice map, main concept: Sheet: J-I 3

Also used in: none

Question 15

How to find the equation of the circle drawn on the chord cut by a given line on a given 2nd degree curve as diameter?

Model Answer

Eliminate y: quadratic in x (roots x_1, x_2); eliminate x: quadratic in y (roots y_1, y_2). Make the leading coefficients equal and add. For a circle and a line use $S + \lambda L = 0$ with the centre on L.

Ex: $y = x$ on $x^2+y^2+10x-12y-40 = 0$: $x^2-x-20 = 0$, $y^2-y-20 = 0$; circle $x^2+y^2-x-y-40 = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 26; J-II 3 | DPP: 8(13)

Also used in: Sheet: J-M 2 | DPP: 3(20) | Question Bank: 57

Intercepts

Question 16

What is the formula for the length of a chord of a circle and how is it derived?

Model Answer

$$\text{chord} = 2\sqrt{r^2 - p^2}$$

p = perpendicular distance of the chord from the centre; the perpendicular bisects the chord (Pythagoras).

Ex: $3x - 4y + 6 = 0$ on $x^2 + y^2 - 2x - 2y - 2 = 0$: $r = 2$, $p = 1$, chord $2\sqrt{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 4; J-II 6-8, 69; J-M 19, 30; C-II 16 | DPP: 2(7); 8(10) | Question Bank: 42, 45, 69

Also used in: Sheet: J-I 20; J-II 22, 71; J-A 20

Question 17
What is the formula for the x-intercept and y-intercept of a circle?

Model Answer
x-intercept $= 2\sqrt{g^2 - c}$, y-intercept $= 2\sqrt{f^2 - c}$ (for $x^2 + y^2 + 2gx + 2fy + c = 0$).
Ex: touches y-axis at $(0, 4)$, x-intercept 6: centre $(\pm 5, 4)$, $r = 5$.

Grade - Medium
Practice map, main concept: Sheet: J-I 6 | DPP: 2(6)
Also used in: Sheet: J-II 2, 63; J-A 4 | DPP: 2(2, 13, 16) | Question Bank: 41

Question 18
What is the general method to find the length of a chord for any 2nd degree curve?

Model Answer
Put $y = mx + c$ in the curve: quadratic in x with roots x_1, x_2; chord $= |x_1 - x_2|\sqrt{1 + m^2} = \frac{\sqrt{D}}{|a|}\sqrt{1 + m^2}$.

Grade - Low
Practice map, main concept: Sheet: C-I 16
Also used in: none

Question 19
What is the general method to find the x-intercept and y-intercept for any 2nd degree curve?

Model Answer
Put $y = 0$ (or $x = 0$): intercept $= |x_1 - x_2| = \frac{\sqrt{D}}{|a|}$ of the resulting quadratic.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 20
What is the condition for a circle to intersect the x-axis / y-axis at 2 distinct points?

Model Answer
$g^2 > c$ (x-axis); $f^2 > c$ (y-axis).

Grade - Medium
Practice map, main concept: Sheet: J-A 7
Also used in: none

Question 21
What is the condition for a circle to touch the x-axis / y-axis?

Model Answer
$g^2 = c$ (x-axis); $f^2 = c$ (y-axis).
Ex: through $(1, -2), (3, -4)$ touching the x-axis: $x^2+y^2-6x+4y+9 = 0$ or $x^2+y^2+10x+20y+25 = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-II 67; J-A 4 | DPP: 2(15)
Also used in: Sheet: J-II 59; J-M 17; J-A 7 | Question Bank: 67

Question 22

What is the condition for a circle neither touching nor intersecting the x-axis / y-axis?

Model Answer

$g^2 < c$ (circle wholly above or below the x-axis); $f^2 < c$ (wholly left or right of the y-axis).

Ex: $x^2 + y^2 - 6x - 10y + c = 0$ meets neither axis and $(1, 4)$ inside: $25 < c < 29$.

Grade - Low

Practice map, main concept: Sheet: J-I 9 | DPP: 2(19)

Also used in: none

Question 23

What is the condition on the coordinates of the centre if the circle touches the x-axis / y-axis?

Model Answer

Touches x-axis: $|k| = r$; touches y-axis: $|h| = r$; touches both: $|h| = |k| = r$, i.e. $(x \mp r)^2 + (y \mp r)^2 = r^2$ (4 circles).

Grade - High

Practice map, main concept: Sheet: J-I 5; J-II 63; J-M 17 | DPP: 2(14, 16) | Question Bank: 41, 67

Also used in: Sheet: J-I 29; J-II 61; J-M 19, 31-32; C-I 7; C-II 17, 27 | DPP: 3(16); 6(5); 7(6); 8(3) | Question Bank: 39, 46-47, 62, 69

Question 24

Draw the diagram of a chord drawn in a circle and find all lengths / area / angle.

Model Answer

Chord AB at distance p, half-angle θ at centre: $AB = 2\sqrt{r^2 - p^2} = 2r\sin\theta$, $p = r\cos\theta$, area $\triangle OAB = \frac{1}{2}r^2 \sin 2\theta = p\sqrt{r^2 - p^2}$; angle at major arc $= \theta$, at minor arc $= \pi - \theta$.

Grade - Low

Practice map, main concept: Sheet: J-II 5

Also used in: none

Parametric Form of a Circle

Question 25

How to assume the coordinates of a point on a given circle?

Model Answer

$(h + r\cos\theta,\ k + r\sin\theta)$, $\theta \in [0, 2\pi)$, for centre (h, k) and radius r; eliminate θ with $\cos^2\theta + \sin^2\theta = 1$ in locus questions.

Ex: P on $x^2 + y^2 - 2x - 2y - 2 = 0$: mid-point of OP lies on $x^2 + y^2 - x - y - \frac{1}{2} = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-M 9, 13; C-II 24 | DPP: 1(11); 2(9, 17) | Question Bank: 54

Also used in: Sheet: J-I 7; C-II 1 | Question Bank: 40

Question 26

What are the parametric coordinates of a point on the standard circle?

Model Answer

$(a\cos\theta,\ a\sin\theta)$ on $x^2+y^2=a^2$

Use for max / min of linear expressions: $\max(px+qy)=ph+qk+r\sqrt{p^2+q^2}$.

Ex: (a,b) on $x^2+y^2-2x-2y-2=0$: $\max(5a-12b)=-7+26=19$.

Grade - Medium

Practice map, main concept: Sheet: J-I 8; J-II 9-10; J-M 29; C-I 11 | DPP: 2(10) | Question Bank: 44

Also used in: none

Position of a Point w.r.t. a Circle

Question 27

How to define S, S1 and T for a point w.r.t. any 2nd degree curve?

Model Answer

S = the curve's expression (RHS 0, coefficient of x^2 made 1 for a circle); $S_1=S(x_1,y_1)$; T: replace $x^2\to xx_1$, $y^2\to yy_1$, $2x\to x+x_1$, $2y\to y+y_1$, $2xy\to xy_1+yx_1$.

$$T\equiv xx_1+yy_1+g(x+x_1)+f(y+y_1)+c$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 28

How to find the relative position of a point w.r.t. a circle?

Model Answer

$S_1>0$: outside; $S_1=0$: on; $S_1<0$: inside (equivalently $CP>r,=r,<r$).

Ex: $(1,2)$: $S_1=10>0$ for $x^2+y^2-4x+6y-3$ (outside), $-6<0$ for $x^2+y^2+4x-6y-3$ (inside).

Grade - Medium

Practice map, main concept: Sheet: J-II 13, 54-55; J-M 16 | DPP: 2(12, 18) | Question Bank: 66

Also used in: Sheet: J-I 9; C-II 21 | DPP: 1(19); 2(19); 5(17)

Question 29

How to find the greatest and least distance of a given point A from a circle?

Model Answer

$$\text{greatest}=AC+r,\quad \text{least}=|AC-r|$$

Ex: $M(-7,2)$, $x^2+y^2-10x-14y-151=0$ (centre $(5,7)$, $r=15$): $AC=13$, least distance 2.

Grade - Medium

Practice map, main concept: Sheet: J-I 7; J-II 11; J-M 31; C-I 19 | Question Bank: 46

Also used in: none

Question 30

How to find the point on a circle lying at the greatest / least distance from a given point A?

Model Answer

Both lie on line AC: $C \pm r\,\hat{u}$ with $\hat{u}$ the unit vector along CA (nearest on the side of A); use the parametric form of line AC.

Ex: farthest point of $x^2 + y^2 - 12x - 4y + 30 = 0$ from O: centre $(6, 2)$, $r = \sqrt{10}$, point $(9, 3)$.

Grade - Low

Practice map, main concept: Sheet: J-II 12

Also used in: none

Question 31

How to find the least distance between a line and a circle?

Model Answer

$p =$ distance of the line from the centre: least distance $= p - r$ if $p > r$, else 0.

Ex: $3x + 4y = 25$ and $x^2 + y^2 = 6x - 8y$ (centre $(3, -4)$, $r = 5$): $p = \frac{32}{5}$, least distance $\frac{7}{5}$.

Grade - Low

Practice map, main concept: DPP: 3(2, 4)

Also used in: Sheet: J-II 18; C-I 9

Line and Circle

Question 32

What is the condition of tangency for a circle and how is it used?

Model Answer

$$p = r$$

Distance of the line from the centre equals the radius. For $y = mx + c$ and $x^2 + y^2 = a^2$: $c^2 = a^2(1 + m^2)$. A circle touching two lines has its centre on an angle bisector.

Ex: $3x + 4y = c$ touches $x^2 + y^2 = 100$ iff $c = \pm 50$.

Grade - High

Practice map, main concept: Sheet: J-I 11; J-II 15, 19, 48-50, 58, 61, 68; J-M 4, 14, 27, 32; J-A 1, 20; C-I 8, 10; C-II 15, 17 | DPP: 1(8, 14); 2(8); 3(1, 15-16, 20) | Question Bank: 47, 55, 59, 62, 64

Also used in: Sheet: J-I 10, 22; J-II 1, 8, 22, 53, 56; J-M 3, 26; C-I 1-2; C-II 7, 12, 19 | DPP: 3(5, 10, 21); 6(16) | Question Bank: 58, 63

Question 33

How to determine the relative position of a line w.r.t. a circle?

Model Answer

$p > r$: no common point; $p = r$: tangent; $p < r$: secant; $p = 0$: diameter.

Ex: $3x + 4y = c$ cuts $x^2 + y^2 = 100$ at two points iff $-50 < c < 50$.

Grade - Medium

Practice map, main concept: Sheet: J-II 18, 56; C-II 21 | DPP: 1(9); 3(3, 5, 17); 7(12)

Also used in: Sheet: J-M 16 | Question Bank: 66

Question 34
What is the general method to find the relative position of a line w.r.t. a 2nd degree curve?

Model Answer
Eliminate one variable to get a quadratic: $D > 0$ secant, $D = 0$ tangent, $D < 0$ no meeting.

Grade - Low
Practice map, main concept: DPP: 2(11)
Also used in: Sheet: C-I 16

Question 35
What is the general method to find the condition of tangency of a line w.r.t. a 2nd degree curve?

Model Answer
Substitute the line in the curve and put the discriminant of the quadratic $= 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Tangent and Normal

Question 36
Define tangent.

Model Answer
Limiting position of a secant when its two points of intersection coincide.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 37
How to find the equation of the tangent to a 2nd degree curve if the point of tangency is given?

Model Answer
$T = 0$: $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
Ex: tangent to $x^2 + y^2 = 5$ at $(1, -2)$: $x - 2y = 5$.

Grade - Medium
Practice map, main concept: Sheet: J-I 12; C-II 14
Also used in: Sheet: J-I 23; J-A 1, 6

Question 38
How to find the equation of a tangent to a 2nd degree curve if any condition other than the point of tangency is given?

Model Answer
Assume the line ($y = mx + c$ with the given slope, or through the given point) and apply the condition of tangency $p = r$. For $x^2 + y^2 = a^2$: $y = mx \pm a\sqrt{1 + m^2}$.
Ex: tangents to $x^2 + y^2 - 6x + 4y = 12$ parallel to $4x + 3y + 5 = 0$: $4x + 3y - 31 = 0$, $4x + 3y + 19 = 0$.

Grade - Low
Practice map, main concept: Sheet: J-I 10; J-II 53 | DPP: 3(10, 18-19, 22)
Also used in: Sheet: J-I 4; J-II 14, 57 | DPP: 3(1)

Question 39
What is the equation of the tangent at a point given in parametric form on the standard circle?

Model Answer

$$x\cos\theta + y\sin\theta = a$$

At $(a\cos\theta, a\sin\theta)$ on $x^2+y^2=a^2$; its intercepts are $a\sec\theta$, $a\,\mathrm{cosec}\,\theta$.
Ex: mid-point of the intercepted part lies on $a^2(x^2+y^2) = 4x^2y^2$.

Grade - Medium
Practice map, main concept: Sheet: J-II 16-17; J-A 10; C-II 25 | DPP: 4(11)
Also used in: Sheet: J-A 6; C-I 13 | DPP: 7(13)

Question 40
How to find the equations of tangents to a circle passing through an external point?

Model Answer
Take $y - y_1 = m(x - x_1)$ and apply $p = r$: quadratic in m gives two tangents (if one root is missing, the other tangent is $x = x_1$). Or use $SS_1 = T^2$ (67).
Ex: from $D(3,-3)$ to $(x+4)^2+(y+2)^2 = 25$: $3x - 4y = 21$ and $4x + 3y = 3$.

Grade - Low
Practice map, main concept: Sheet: J-II 20, 70 | DPP: 3(6, 13); 4(10)
Also used in: none

Question 41
Define normal.

Model Answer
The line through the point of contact perpendicular to the tangent.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 42
How to find the equation of a normal to a circle?

Model Answer
Every normal passes through the centre: normal at P = line CP; two given normals meet at the centre.
Ex: normal to $x^2+y^2 = 25$ through $(2,1)$: $x = 2y$.

Grade - Medium
Practice map, main concept: Sheet: J-II 51; J-A 6 | DPP: 3(8-9, 11-12, 14); 6(19)
Also used in: Sheet: J-M 14 | DPP: 6(3) | Question Bank: 55

Question 43
What is the equation of the chord joining two points $P(\alpha)$, $Q(\beta)$ on the standard circle?

Model Answer

$$x\cos\frac{\alpha+\beta}{2} + y\sin\frac{\alpha+\beta}{2} = a\cos\frac{\alpha-\beta}{2}$$

Grade - Low
Practice map, main concept: none
Also used in: none

To Find the Point of Tangency

Question 44

How to find the point of tangency on a circle if the equation of the tangent is given?

Model Answer

Foot of the perpendicular from the centre to the tangent; or compare $T = 0$ (point (x_1, y_1)) with the given line.

Ex: $5x + 12y = 65$ on $x^2 + y^2 = 25$: $\frac{x_1}{5} = \frac{y_1}{12} = \frac{25}{65}$, point $(\frac{25}{13}, \frac{60}{13})$.

Grade - Medium

Practice map, main concept: Sheet: J-II 52; J-M 26 | DPP: 3(7, 21) | Question Bank: 63

Also used in: Sheet: J-I 12; J-A 20; C-II 24 | DPP: 3(15)

Question 45

How to find the point of tangency on a 2nd degree curve if the equation of the tangent is given?

Model Answer

Write the tangent at (x_1, y_1) as $T = 0$ and compare coefficients with the given line (or solve line and curve: the double root).

Grade - Low

Practice map, main concept: none

Also used in: none

Length of Tangent

Question 46

Write the expression for the length of the tangent from an external point.

Model Answer

$$L = \sqrt{S_1} = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$$

Coefficients of x^2, y^2 must be 1 first.

Ex: from $(0, 5)$ to $x^2 + y^2 + 2x - 4 = 0$: $L = \sqrt{21}$; to $x^2 + y^2 - y + 1 = 0$: $L = \sqrt{21}$.

Grade - Low

Practice map, main concept: Sheet: J-I 13-15; J-II 62 | DPP: 4(2, 7, 14); 6(13)

Also used in: Sheet: J-II 20; C-II 13

Question 47

Define chord of contact for an external point.

Model Answer

The chord joining the points of contact of the two tangents from the point; it exists only if the point is not inside the circle.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 48

How to find the angle between the tangents drawn from an external point to a circle?

Model Answer

If the angle between the tangents is 2θ: $\tan\theta = \dfrac{r}{L}$, $\sin\theta = \dfrac{r}{CP}$, so the angle $= 2\tan^{-1}\dfrac{r}{L}$.

Ex: from O to $(x-7)^2+(y+1)^2=25$: $CP=\sqrt{50}$, $\sin\theta=\frac{1}{\sqrt{2}}$, angle $\frac{\pi}{2}$.

Grade - Low

Practice map, main concept: Sheet: J-I 17; C-II 13 | DPP: 4(9, 13); 5(14)

Also used in: none

Question 49

What is the angle subtended by a circle at an external point?

Model Answer

The angle between the two tangents from the point: $2\sin^{-1}\dfrac{r}{CP}$. Two circles subtend equal angles at P iff $\frac{CP_1}{r_1}=\frac{CP_2}{r_2}$ (locus: a circle).

Grade - Low

Practice map, main concept: Sheet: C-I 14 | DPP: 6(12)

Also used in: none

Question 50

How to find the equation of the circumcircle of the triangle formed by the tangents from an external point and its chord of contact?

Model Answer

$PACB$ is cyclic with CP as diameter: $(x-x_1)(x+g)+(y-y_1)(y+f)=0$.

Ex: $P(1,8)$, $x^2+y^2-6x-4y-11=0$: $(x-1)(x-3)+(y-8)(y-2)=0$, i.e. $x^2+y^2-4x-10y+19=0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 16; J-II 26; J-M 11 | DPP: 4(4, 12)

Also used in: Sheet: J-A 13 | DPP: 8(20)

Question 51

Draw the diagram of tangents drawn from an external point to a circle and find all lengths / area / angle.

Model Answer

$L=\sqrt{S_1}$, $R=$ radius: area $PACB=RL$; chord of contact $AB=\dfrac{2RL}{\sqrt{R^2+L^2}}$; area $\triangle PAB=\dfrac{RL^3}{R^2+L^2}$; angle between tangents $\tan^{-1}\dfrac{2RL}{L^2-R^2}$.

Ex: $x^2+y^2=25$, $P(10,0)$: $L=5\sqrt{3}$, $PACB=25\sqrt{3}$, $AB=5\sqrt{3}$, $\triangle PAB=\frac{75\sqrt{3}}{4}$, angle $60°$.

Grade - Medium

Practice map, main concept: Sheet: J-M 10; C-I 9, 24; C-II 1 | DPP: 4(5-6, 8)

Also used in: Sheet: J-II 20, 62

Power of a Point

Question 52
Define the power of a point w.r.t. a circle and how to find it.

Model Answer

$$\text{Power} = S_1 = PT^2 = PA \cdot PB$$

Constant for every line through P meeting the circle at A, B (signed lengths along the line).
Ex: $P(5,2)$, $x^2+y^2=25$: $PA \cdot PB = 4$ for every secant.

Grade - Low
Practice map, main concept: Sheet: J-II 21; C-I 15; C-II 3 | DPP: 4(1, 3)
Also used in: Sheet: J-II 45, 62; C-I 24

Question 53
What is the power of a point according to its position relative to the circle?

Model Answer
Positive outside, zero on, negative inside the circle.

Grade - Low
Practice map, main concept: none
Also used in: none

Director Circle

Question 54
Define the director circle of a curve.

Model Answer
Locus of the point of intersection of two perpendicular tangents to the curve.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 55
What is the director circle of a circle?

Model Answer
$x^2+y^2=2a^2$ for $x^2+y^2=a^2$
Concentric, radius $\sqrt{2}\,r$. Ex: for $(x-2)^2+(y+1)^2=2$: $(x-2)^2+(y+1)^2=4$.

Grade - Low
Practice map, main concept: Sheet: C-II 4, 23
Also used in: Sheet: C-II 9

Question 56
What is the general method to find the director circle of a 2nd degree curve?

Model Answer
Write the tangent of slope m (condition of tangency), pass it through (h,k): quadratic in m; put $m_1 m_2 = -1$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 57
How to find the locus of an external point P for which the angle between the tangents from P to the circle is constant?

Model Answer
Angle 2θ: $CP = r\,\mathrm{cosec}\,\theta$, a concentric circle.

Ex: angle 60° for $x^2 + y^2 = a^2$: $x^2 + y^2 = 4a^2$.

Grade - Low
Practice map, main concept: none
Also used in: none

Chord whose Mid-point is Given

Question 58
Write the equation of a chord of a 2nd degree curve whose mid-point is given.

Model Answer

$$T = S_1$$

Locus of mid-points: write $T = S_1$ at (h, k) and apply the given condition. Ex: chord of $x^2 + y^2 - 11x - 2y - 24 = 0$ bisected at $(1, 2)$: $9x - 2y - 5 = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-II 23-24; J-M 6; J-A 3; C-I 5 | DPP: 5(3-5, 8-9, 11, 15, 17, 19) | Question Bank: 61
Also used in: DPP: 5(18)

Question 59
What is the formula for the length of a chord of a circle whose mid-point is given?

Model Answer

$$\text{length} = 2\sqrt{-S_1} = 2\sqrt{r^2 - CM^2}$$

Ex: circle $x^2 + y^2 - 4x - 6y + 8 = 0$ ($r^2 = 5$), mid-point $(1, 2)$: $CM^2 = 2$, chord $2\sqrt{3}$.

Grade - Medium
Practice map, main concept: Question Bank: 43
Also used in: none

Question 60
How to find the mid-point of a chord of a circle whose equation is given?

Model Answer
Foot of the perpendicular from the centre to the chord (the diameter perpendicular to the chord bisects it).

Ex: $2x - 5y + 18 = 0$ on $x^2 + y^2 - 6x + 2y - 54 = 0$: mid-point $(1, 4)$.

Grade - Medium
Practice map, main concept: Sheet: J-I 18-19; J-A 11 | DPP: 5(2, 16, 18)
Also used in: Question Bank: 42

Question 61

What is the general method to find the mid-point of a chord, whose equation is given, of a 2nd degree curve?

Model Answer

Take mid-point (x_1, y_1), write $T = S_1$ and compare with the given chord; or solve line and curve and take $\frac{x_1+x_2}{2}$ from the sum of roots.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 62

Which is the shortest chord of a circle passing through an internal point?

Model Answer

The chord bisected at that point (perpendicular to CM), length $2\sqrt{r^2 - CM^2}$; it is the chord through M farthest from the centre. The longest is the diameter through M.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 63

What is the locus of the mid-point of a variable chord of a circle passing through a fixed point P?

Model Answer

$\angle CMP = 90°$: the circle on CP as diameter (the part inside the given circle).

Ex: chords of $x^2 + y^2 - 4x = 0$ through $(3, 4)$: $x^2 + y^2 - 5x - 4y + 6 = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-M 25 | DPP: 5(1) | Question Bank: 52

Also used in: none

Chord of Contact

Question 64

Write the equation of the chord of contact w.r.t. a 2nd degree curve for an external point.

Model Answer

$$T = 0$$

Ex: from $(8, 3)$ to $x^2 + y^2 = 49$: $8x + 3y = 49$.

Grade - Medium

Practice map, main concept: Sheet: J-I 22; C-II 19 | DPP: 5(12)

Also used in: Sheet: J-II 26; J-A 3 | DPP: 4(8); 5(15)

Question 65
How to find the external point if the equation of the chord of contact is given?

Model Answer
Write $T = 0$ for unknown (x_1, y_1) and compare coefficients with the given line (the point is the pole of the line). The tangents at the ends of a chord meet at this point.

Ex: tangents at the ends of $x - 2y + 1 = 0$ on $x^2 + y^2 = 25$ meet at $(-25, 50)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 23; J-II 27; J-M 8 | DPP: 5(20); 7(13-14)
Also used in: Sheet: J-I 25; J-II 25; J-A 9 | DPP: 8(15)

Question 66
What is the point of intersection of the tangents drawn at $P(\alpha)$, $Q(\beta)$ on the standard circle?

Model Answer

$$\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}} \right)$$

Grade - Medium

Practice map, main concept: Sheet: J-M 3, 7; J-A 9 | Question Bank: 58
Also used in: none

Pair of Tangents

Question 67
Write the combined equation of the pair of tangents from an external point to a 2nd degree curve.

Model Answer

$$SS_1 = T^2$$

Ex: from the origin to $x^2 + y^2 + 2gx + 2fy + c = 0$: $(gx + fy)^2 = c(x^2 + y^2)$.

Grade - Low

Practice map, main concept: Sheet: J-I 24-25
Also used in: Sheet: J-II 70

Family of Circles: Type 1

Question 68
How to find the equation of a circle passing through the points of intersection of two circles S1 and S2?

Model Answer
$S_1 + \lambda S_2 = 0$ $(\lambda \neq -1)$, or $S_1 + \lambda(S_1 - S_2) = 0$
Find λ from the extra condition. Ex: through the meet of $x^2 + y^2 - 6x + 2y + 4 = 0$, $x^2 + y^2 + 2x - 4y - 6 = 0$ with centre on $y = x$: $7x^2 + 7y^2 - 10x - 10y - 12 = 0$.

Grade - Low

Practice map, main concept: Sheet: J-I 35; J-II 32, 40, 46; C-I 3 | DPP: 6(15, 17-18); 8(20)
Also used in: none

Question 69
How to check whether the value of λ is infinite by observing the calculations, and how to find the required circle in that case?

Model Answer
If the condition gives an impossible equation in λ (e.g. $0 = k$) or is already satisfied by S_2, then $\lambda \to \infty$: the answer is $S_2 = 0$ itself. Writing $\mu S_1 + S_2 = 0$ avoids the loss.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 35

Family of Circles: Type 2

Question 70
How to find the equation of a circle passing through the points of intersection of a circle S and a line L?

Model Answer

$$S + \lambda L = 0$$

Ex: through $(3, 4)$ and the meet of $x^2 + y^2 = 9$, $x + y = 3$: $\lambda = -4$, $x^2 + y^2 - 4x - 4y + 3 = 0$.

Grade - Low
Practice map, main concept: DPP: 6(16)
Also used in: Sheet: J-I 26

Question 71
What does the equation of a variable circle $S + \lambda L = 0$ represent?

Model Answer
Circles through the two fixed points where $S = 0$ meets $L = 0$. To find them, group the λ terms: solve $S = 0$, $L = 0$.
Ex: $x^2 + y^2 + kx + (1 + k)y - (k + 1) = 0$ passes through $(1, 0)$ and $(\frac{1}{2}, \frac{1}{2})$.

Grade - Low
Practice map, main concept: Sheet: J-II 60; C-II 10 | DPP: 8(14)
Also used in: Sheet: J-II 41 | DPP: 4(3)

Family of Circles: Type 3

Question 72
How to find the equation of the family of circles passing through two given points A and B?

Model Answer

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + \lambda \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Circle on AB as diameter $+\lambda$ (line AB). Centres lie on the perpendicular bisector of AB.

Grade - Low
Practice map, main concept: Sheet: J-II 33, 59
Also used in: Sheet: J-II 67; C-II 8 | DPP: 2(15)

Family of Circles: Type 4

Question 73
How to find the equation of a circle touching a given line L at its given point P?

Model Answer

$$(x-x_1)^2+(y-y_1)^2+\lambda L=0$$

Ex: through O, touching $x+y=2$ at $(1,1)$: $\lambda=1$, $x^2+y^2-x-y=0$.

Grade - Low
Practice map, main concept: Sheet: J-I 27-28; J-II 29, 44, 47 | DPP: 6(1-3); 8(4)
Also used in: Sheet: C-II 9

Question 74
How to find the equation of a circle touching a given 2nd degree curve at its given point P?

Model Answer
Use the tangent at P $(T=0)$: $(x-x_1)^2+(y-y_1)^2+\lambda T=0$; for a given circle S also $S+\lambda T=0$.
Ex: through $(1,1)$, touching $x^2+y^2+4x-6y-3=0$ at $(2,3)$: $x^2+y^2+x-6y+3=0$.

Grade - Low
Practice map, main concept: Sheet: J-II 30
Also used in: Sheet: J-II 31

Question 75
How to find the equation of the circumcircle of the triangle formed by three lines whose equations are given?

Model Answer

$$L_1L_2+\lambda L_2L_3+\mu L_3L_1=0$$

Choose λ,μ so that coeff of x^2 = coeff of y^2 and coeff of $xy=0$. Ex: $x=0$, $y=0$, $x+y=4$: $x^2+y^2-4x-4y=0$.

Grade - Low
Practice map, main concept: DPP: 1(21)
Also used in: none

Question 76
How to find the equation of the circumcircle of a cyclic quadrilateral formed by 4 lines whose equations are given?

Model Answer

$$L_1L_3+\lambda L_2L_4=0$$

Sides in order L_1,L_2,L_3,L_4; fix λ by coeff of x^2 = coeff of y^2, coeff of $xy=0$. Ex: $2x+3y=2$, $3x-2y=4$, $x+2y=3$, $2x-y=3$ in this order: $8x^2+8y^2-25x-3y+18=0$.

Grade - Low
Practice map, main concept: Sheet: J-II 1
Also used in: none

Common Tangents

Question 77
Define common tangents and their types DCT and TCT.

Model Answer
A line touching both circles. Direct common tangent (DCT): both circles on the same side; transverse common tangent (TCT): circles on opposite sides.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 78
Which common tangent is called external and which one internal?

Model Answer
DCT = external common tangent; TCT = internal (transverse) common tangent (it passes between the circles).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 79
How to check whether a common tangent is a DCT or a TCT?

Model Answer
Put both centres in the line L: same sign $\Rightarrow$ DCT, opposite signs $\Rightarrow$ TCT.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 80
What is the relative position of the centres of two circles w.r.t. a DCT / TCT?

Model Answer
Both centres on the same side of a DCT; on opposite sides of a TCT.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 81
In what ratio does the point of intersection of the DCTs / TCTs divide the line joining the centres?

Model Answer
DCTs meet at P dividing C_1C_2 externally in $r_1 : r_2$; TCTs meet at R dividing it internally in $r_1 : r_2$. Ex: $C_1C_2 = 35$, $r_1 = 12$, $r_2 = 9$: $C_1R = 35 \cdot \frac{12}{21} = 20$.

Grade - Medium
Practice map, main concept: Sheet: J-A 12 | DPP: 6(6); 7(1, 8)
Also used in: Sheet: J-M 35 | DPP: 7(15) | Question Bank: 50

Question 82
What is the condition on the DCTs if the radii of the two circles are equal?

Model Answer
They are parallel to the line joining the centres (no finite meeting point).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 83
What is the formula for the length of a DCT / TCT?

Model Answer

$$L_{DCT} = \sqrt{d^2 - (r_1 - r_2)^2}, \qquad L_{TCT} = \sqrt{d^2 - (r_1 + r_2)^2}$$

Ex: $x^2 + y^2 - 20x + 64 = 0$, $x^2 + y^2 + 30x + 144 = 0$: $d = 25$, $r_1 = 6$, $r_2 = 9$: $L_{TCT} = 20$.

Grade - Medium
Practice map, main concept: Sheet: J-II 34 | DPP: 6(9, 11)
Also used in: Sheet: J-II 65, 71; J-A 12, 19; C-I 23

Question 84
How to find the equations of the DCTs / TCTs of two circles?

Model Answer
Find P (external) / R (internal division in $r_1 : r_2$), then tangents from that point to either circle ($y - k = m(x - h)$, $p = r$). If $r_1 = r_2$, the DCTs are $y = mx + c$ with $m =$ slope of C_1C_2.
Ex: $x^2+y^2 = 1$, $(x-1)^2+(y-3)^2 = 4$: common tangents $y = 1$, $x+1 = 0$, $3x+4y = 5$, $4x-3y = 5$.

Grade - Medium
Practice map, main concept: Sheet: J-A 2; C-I 1 | DPP: 6(10); 7(18)
Also used in: Sheet: C-II 16

Question 85
How to find the equation of the DCT / TCT if two circles touch each other?

Model Answer
The common tangent at the point of contact is $S_1 - S_2 = 0$; the point of contact divides C_1C_2 in $r_1 : r_2$ (internally if they touch externally, externally if internally).
Ex: $x^2 + y^2 - 4x - 6y - 12 = 0$, $x^2 + y^2 + 6x + 18y + 26 = 0$: $5x + 12y + 19 = 0$.

Grade - Low
Practice map, main concept: DPP: 7(15)
Also used in: none

Question 86
How to find the angle between the DCTs / TCTs?

Model Answer
Half-angle θ: $\sin\theta = \dfrac{r_1 - r_2}{d}$ for DCTs, $\sin\theta = \dfrac{r_1 + r_2}{d}$ for TCTs; the angle is 2θ.
Ex: radii 4 and 1 touching externally ($d = 5$): $\sin\theta = \frac{3}{5}$, $\sin 2\theta = \frac{24}{25}$.

Grade - Low
Practice map, main concept: DPP: 6(4)
Also used in: none

Position of Two Circles relative to Each Other

Question 87

When are two circles separated from each other, and how many common tangents are there?

Model Answer

$d > r_1 + r_2$: 4 common tangents (2 DCT, 2 TCT).

Grade - Medium

Practice map, main concept: Sheet: J-II 65, 72 | DPP: 7(5)

Also used in: Sheet: J-A 15-16

Question 88

When do two circles touch each other externally, and how many common tangents are there?

Model Answer

$d = r_1 + r_2$: 3 common tangents. A circle touching a fixed circle externally and a fixed line: distance to centre $= r + R$, distance to line $= r$ (locus: parabola).

Ex: $x^2 + y^2 - 2x - 4y + 1 = 0$, $x^2 + y^2 + 4x + 4y - 1 = 0$: $d = 5 = 2 + 3$, touch externally.

Grade - Medium

Practice map, main concept: Sheet: J-I 29, 31, 37; J-II 2, 36, 64, 66; J-M 15, 35; J-A 18; C-I 7; C-II 5, 27 | DPP: 6(5, 7); 7(7, 10-11, 16-17) | Question Bank: 39, 50

Also used in: Sheet: J-II 72; C-I 10, 21 | DPP: 3(11); 6(19)

Question 89

When do two circles intersect at 2 distinct points, and how many common tangents are there?

Model Answer

$|r_1 - r_2| < d < r_1 + r_2$: 2 common tangents (DCTs only).

Ex: $(x + 1)^2 + (y + 2)^2 = r^2$ and $x^2 + y^2 - 4x - 4y + 4 = 0$ ($d = 5$, $r_2 = 2$): $3 < r < 7$.

Grade - Medium

Practice map, main concept: Sheet: J-II 35; J-M 21, 24; C-I 23 | DPP: 7(2)

Also used in: Sheet: J-II 72

Question 90

When does one circle touch another internally, and how many common tangents are there?

Model Answer

$d = |r_1 - r_2|$: 1 common tangent.

Grade - Medium

Practice map, main concept: Sheet: J-I 30; J-II 31; J-M 37; J-A 17; C-I 21; C-II 12 | DPP: 6(8); 7(6)

Also used in: Sheet: J-II 72; C-II 27

Question 91

When is one circle contained in another, and how many common tangents are there?

Model Answer

$d < |r_1 - r_2|$: no common tangent.

Grade - Medium

Practice map, main concept: Sheet: J-A 15-16

Also used in: DPP: 7(5)

Question 92
What is the minimum distance between two circles in every relative position?

Model Answer
Separate: $d-(r_1+r_2)$; touching or intersecting: 0; one inside the other: $|r_1-r_2|-d$. Maximum distance: $d+r_1+r_2$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 6(11)

Radical Axis

Question 93
Define the radical axis of two circles.

Model Answer
Locus of a point whose powers (squares of tangent lengths) w.r.t. the two circles are equal.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 94
Write the expression for the radical axis of two circles S1 = 0 and S2 = 0.

Model Answer

$$S_1 - S_2 = 0: \ 2(g_1-g_2)x + 2(f_1-f_2)y + c_1 - c_2 = 0$$

Coefficients of x^2, y^2 must be 1 in both.

Grade - Medium
Practice map, main concept: Sheet: J-II 37; C-II 7 | DPP: 7(3)
Also used in: Sheet: J-I 32-33; J-II 39; J-A 19 | DPP: 7(14)

Question 95
Which line is the radical axis of two circles if they intersect at 2 distinct points?

Model Answer
Their common chord.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 37 | DPP: 7(3)

Question 96
Which line is the radical axis of two circles if they touch each other?

Model Answer
The common tangent at the point of contact.

Grade - Low
Practice map, main concept: Sheet: J-I 32
Also used in: Sheet: J-II 64, 66

Question 97
What is the condition on the radical axis w.r.t. the line joining the centres?

Model Answer
It is perpendicular to C_1C_2 (but need not pass through its mid-point).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 98
What is the condition on the radical axis w.r.t. a common tangent of two circles?

Model Answer
It bisects every common tangent (equal tangent lengths from the meeting point).

Grade - Medium
Practice map, main concept: Sheet: J-A 8, 19
Also used in: Sheet: J-II 34

Question 99
Which type of two circles will not have a radical axis?

Model Answer
Concentric circles ($S_1 - S_2 = c_1 - c_2 \neq 0$ is not a line).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 100
How to find the equation of the common chord of two circles?

Model Answer

$$S_1 - S_2 = 0$$

Length $= 2\sqrt{r_1^2 - p_1^2}$, $p_1 =$ distance of $S_1 - S_2 = 0$ from C_1; circle on it as diameter: $S_1 + \lambda(S_1 - S_2) = 0$ with centre on the chord.
Ex: radii 4 and 8 cutting orthogonally: $d = 4\sqrt{5}$, common chord $= \frac{2r_1r_2}{d} = \frac{16}{\sqrt{5}}$.

Grade - Medium
Practice map, main concept: Sheet: J-II 71; J-M 33; C-II 20 | DPP: 7(9) | Question Bank: 48
Also used in: Sheet: J-M 28; C-I 3 | DPP: 6(17); 7(4); 8(6) | Question Bank: 65

Question 101
What is the condition if one circle bisects the circumference of the other circle?

Model Answer
If S_2 bisects the circumference of S_1, the common chord $S_1 - S_2 = 0$ passes through the centre of S_1 (it is a diameter of S_1): $r_2^2 = r_1^2 + C_1C_2^2$.
Ex: $x^2 + y^2 - 2x + 3y + c = 0$ bisects $x^2 + y^2 = 9$: $c = -9$.

Grade - Medium
Practice map, main concept: Sheet: J-I 33-34, 36; J-II 38; J-M 22 | DPP: 6(14); 7(4)
Also used in: DPP: 1(15)

Question 102
How to check the relative position of two circles with the help of the radical axis?

Model Answer
Compare the distance p of $S_1 - S_2 = 0$ from C_1 with r_1: $p < r_1$ intersect, $p = r_1$ touch, $p > r_1$ no common point (separate or one inside the other; the axis then lies outside both).

Grade - Low
Practice map, main concept: none
Also used in: none

Radical Centre

Question 103
Define the radical centre of three circles.

Model Answer
The common point of the three radical axes taken in pairs (exists if the centres are not collinear).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 104
How to find the radical centre of three circles if their equations are given?

Model Answer
Solve any two of $S_1 - S_2 = 0$, $S_2 - S_3 = 0$ (coefficients of x^2, y^2 unity).
Ex: $x^2 + y^2 - 3x - 6y + 14 = 0$, $x^2 + y^2 - x - 4y + 8 = 0$, $x^2 + y^2 + 2x - 6y + 9 = 0$: radical centre $(1, 2)$.

Grade - Low
Practice map, main concept: Sheet: C-II 8 | DPP: 1(15); 8(17)
Also used in: none

Question 105
Which is the point from which the tangents drawn to the three circles are equal?

Model Answer
The radical centre.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 106
Which point is the radical centre of three circles drawn on the three sides of a triangle as diameters?

Model Answer
The orthocentre of the triangle (each radical axis is an altitude).

Grade - Low
Practice map, main concept: none
Also used in: none

Coaxial System of Circles

Question 107
Define a coaxial system of circles.

Model Answer
A family of circles every two of which have the same radical axis.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 108
How to find the equation of a circle belonging to a given coaxial system?

Model Answer
$S_1 + \lambda(S_1 - S_2) = 0$, i.e. $S + \lambda L = 0$ (L = radical axis)
Ex: through O, coaxial with $x^2+y^2 = 9$ and $x^2+y^2-6x+3 = 0$: $x^2+y^2-9+3(x^2+y^2-6x+3) = 0$, i.e. $2x^2 + 2y^2 - 9x = 0$.

Grade - Low
Practice map, main concept: Sheet: J-II 39; C-II 9
Also used in: none

Angle between a Line and a Circle

Question 109
Define the angle of intersection of two curves.

Model Answer
The angle between their tangents at a common point.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 110
What is the angle of intersection of two curves if they have no common point?

Model Answer
Not defined.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 111
How to find the angle between a line and a circle?

Model Answer

$$\cos\theta = \frac{p}{r}$$

p = distance of the line from the centre ($\theta = 0$ for a tangent). Ex: $x^2 + y^2 = 2$ and $3x + 4y = 5$: $p = 1$, $\cos\theta = \frac{1}{\sqrt{2}}$, $\theta = 45°$.

Grade - Low
Practice map, main concept: DPP: 8(1, 12)
Also used in: none

Question 112
What do you mean when a line cuts a circle orthogonally?

Model Answer
$\theta = 90°$: $p = 0$, the line passes through the centre (it is a normal). A line pair cuts a circle orthogonally iff both lines pass through the centre.

Grade - Low
Practice map, main concept: Sheet: J-II 42
Also used in: Sheet: J-II 43 | DPP: 8(5)

Angle between Two Circles and Orthogonality

Question 113
How to find the angle between two circles?

Model Answer

$$\cos\theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2}$$

Ex: $(x-1)^2 + y^2 = 10$, $x^2 + (y-2)^2 = 5$: $d^2 = 5$, $\cos\theta = \frac{10}{2\sqrt{50}} = \frac{1}{\sqrt{2}}$, $\theta = \frac{\pi}{4}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 28 | DPP: 8(9) | Question Bank: 65
Also used in: none

Question 114
What is the condition of orthogonality of two circles in terms of their radii and the distance between centres?

Model Answer

$$r_1^2 + r_2^2 = d^2$$

Ex: equal circles centred $(2,3)$, $(5,6)$ cutting orthogonally: $2r^2 = 18$, $r = 3$.

Grade - Low
Practice map, main concept: DPP: 8(2-3, 6)
Also used in: none

Question 115
What is the condition of orthogonality of two circles in terms of their equations?

Model Answer

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

Equivalent: tangent to one at a common point passes through the other's centre; the power of C_1 w.r.t. S_2 is r_1^2.
Ex: $x^2 + y^2 - 2x - 4 = 0$ and $x^2 + y^2 + 16x + 14y - 12 = 0$: $2(-1)(8) + 0 = -16 = c_1 + c_2$.

Grade - Medium
Practice map, main concept: Sheet: J-I 39; J-II 41, 43, 45; J-A 5; C-I 2 | DPP: 8(5, 7, 16, 18)
Also used in: Sheet: J-II 40, 42, 44, 46-47 | DPP: 8(4, 8)

Question 116
How to find the equation of a circle cutting three given circles orthogonally?

Model Answer
Centre = radical centre of the three circles; radius = tangent length from it to any of them (or solve three orthogonality equations for g, f, c).
Ex: $x^2+y^2-2x+3y-7=0$, $x^2+y^2+5x-5y+9=0$, $x^2+y^2+7x-9y+29=0$: $x^2+y^2-16x-18y-4=0$.

Grade - Low
Practice map, main concept: Sheet: J-I 38 | DPP: 8(19)
Also used in: none

Question 117
What is the locus of the centre of a variable circle cutting two given circles orthogonally?

Model Answer
The radical axis of the two circles (part outside them).
Ex: $x^2+y^2+4x-6y+9=0$, $x^2+y^2-5x+4y+2=0$: $9x-10y+7=0$.

Grade - Medium
Practice map, main concept: Sheet: C-II 11 | DPP: 8(8)
Also used in: Sheet: J-A 5 | DPP: 8(7)

Pole and Polar

Question 118
Define pole and polar for a 2nd degree curve.

Model Answer
Through P draw a variable line meeting the curve at Q, R; the locus of the meet of the tangents at Q, R is the polar of P, and P is the pole of that line.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 119
Write the equation of the polar of a given point.

Model Answer

$$T = 0$$

Conjugate points: each lies on the polar of the other. Ex: $(2, k)$, $(k, 3)$ conjugate w.r.t. $x^2+y^2=10$: $2k+3k=10$, $k=2$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-II 3

Question 120
What does T = 0 represent for a point, according to its position relative to the circle?

Model Answer
Outside: chord of contact (= polar); on the circle: tangent at the point; inside: polar only (no chord of contact), a line not meeting the circle.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 121
How to find the coordinates of the pole of a given line?

Model Answer
Write $T = 0$ for (x_1, y_1) and compare with the line. For $lx + my = n$ and $x^2 + y^2 = a^2$: pole $\left(\frac{a^2 l}{n}, \frac{a^2 m}{n}\right)$.

Grade - Medium
Practice map, main concept: Sheet: J-A 13
Also used in: none

Question 122
From any variable point on a fixed line, tangents are drawn to a 2nd degree curve; the chord of contact passes through which point?

Model Answer
Through the pole of the fixed line.
Ex: points on $2x + y = 4$, circle $x^2 + y^2 = 1$: chords of contact pass through $(\frac{1}{2}, \frac{1}{4})$.

Grade - Low
Practice map, main concept: Sheet: J-II 28
Also used in: none

Question 123
What is the locus of the harmonic conjugate of a point P w.r.t. the 2 points in which any variable line through P cuts a 2nd degree curve?

Model Answer
The polar of P: $T = 0$ (for $x^2 + y^2 = a^2$: $xx_1 + yy_1 = a^2$).

Grade - Low
Practice map, main concept: none
Also used in: none